Scheduling and Routing of Mobile Charging Stations With Stochastic Travel Times to Service Heterogeneous Spatiotemporal Electric Vehicle Charging Requests With Time Windows

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***Abstract*—This article presents an alternative service of mobile charging stations for the large-scale charging of electric vehicles, which consider the spatiotemporal heterogeneity of charging requests. As charging infrastructure is the key determinant for the large-scale adoption of electric vehicles, state-of-the-art scheduling and control strategies need to be explored. The charging of electric vehicles in a conventional charging station even with the fast dc–dc chargers takes around 30 min, which results in congestion and large waiting queues at public charging stations. To account for this issue, a novel strategy of routing and scheduling mobile charging stations to charge electric vehicles without the constraints of time and space is discussed in detail. Furthermore, the traveling times of mobile charging stations in reality are stochastic in nature. We formulate the optimization problem to minimize the cost of charging and show that the problem formulated is a combination of a bin packing problem and a multicity traveling salesman problem; hence, it is NP-hard and cannot be solved in reasonable CPU time, unless P = NP. We, thus, present modified saving's heuristic and modified genetic algorithm metaheuristic to solve the optimization problem. Furthermore, numerical simulations show that the proposed scheduling and routing algorithm requires less number of mobile charging stations and can appreciably reduce the cost of charging.**

***Index Terms*—Charging station, electric vehicle (EV), routing, scheduling.**

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I. Introduction

ELECTRIC vehicles (EVs) being twice as energy efficient as conventional vehicles are fast replacing internal combustion engine-powered vehicles. In order to reduce the greenhouse gas emissions and achieve the regulatory and strategic emission targets, countries are incentivizing the adoption of EVs [2], [3]. According to the International Energy Agency, the number of electric vehicles to hit the road is expected to reach 230 million by 2030 [4].

In spite of so many advantages of adopting EVs, there are challenges that need to be addressed before EVs can be adopted on large scale. Though there has been huge advancement in power electronics, electric motors, and battery technology, the main barrier to the widespread adoption of EVs is the charging infrastructure. Commercial charging stations equipped with fast level-3 chargers have alleviated the problem of charging to some extent.

The chargers have evolved from slow level-1 to fast level-2 ac chargers and now to the faster level-3 dc–dc chargers that deliver about 50–200 kW of power and can, thus, completely charge an EV in around 30 min [27]. Still, the time required to charge an EV is considerably more than refueling conventional gasoline-based vehicles. Large charging times will have three consequences in the near future as the penetration of electric mobility increases.

1) There will be congestion at public charging stations and longer waiting times rendering EVs immobile for longer durations.
2) The peak of EV charging load coinciding with the existing peak demand on the power grid will result in further larger peak and call for system augmentation.
3) Finally, EV owners will face the problem of range anxiety [14], i.e., the fear that the EV will not have sufficient charge to complete a trip and may need to recharge in-between a trip.

A. Motivation

As the charging of the EV takes significant time resulting in congestion and range anxiety, an alternate charging scheme of a

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mobile charging service [1] can alleviate both the problems. Mobile charging stations (MCSs) are vehicles carrying fast dc–dc chargers and battery banks. The main advantage of MCSs is that they can charge EVs without the barrier of time and location, such as at home, office, parking lots, and in-between trips. However, to service a large number of charging requests, the fleet of MCSs need to be properly scheduled and routed to minimize the cost of charging and the number of MCSs and reduce the miss ratio. In densely populated cities, such as Mumbai, Delhi, and Beijing, where it can be difficult to install in-home (residential) chargers as the car owners do not have private parking space and the waiting times at public charging stations are large, a mobile charging service provides an alternate mode of charging. The MCSs can charge EVs at public parking, offices, and residential parking without the constraints of location and time and, thus, help alleviate the problems of congestion and range anxiety.

The EV charging requests are spatially and temporally heterogeneous in nature demanding a particular amount of energy within a preferable time window. Furthermore, the travel times of the MCSs are stochastic in nature as they have to operate on the traffic network, which has different levels of congestion at various intervals of day. Keeping these things in mind, a complete optimization model has been formulated in this article to schedule and route MCSs to service charging requests while satisfying the constraints and minimizing the cost. The optimization problem formulated has been shown to be a combination of a bin packing problem and a multicity traveling salesman problem, i.e., two NP-hard problems. Thus, the overall problem at hand is a combinatorial optimization problem that cannot be solved in reasonable CPU time unless $P = NP$.

B. Contribution

In summary, the contributions of our work are as follows.

1) We model the operation process of a mobile charging service considering stochastic travel times of MCSs in the presence of spatially and temporally heterogeneous charging requests.
2) We formulate the complete optimization problem of scheduling and routing the MCSs to charge EVs while minimizing the cost and reducing the number of MCSs.
3) Finally, we present three different methods for solving the NP-hard combinatorial optimization problem, including exact, heuristic, and metaheuristic methods.

C. Literature Survey

To address the problem of large-scale EV charging, a number of scheduling policies and algorithms have been proposed for both the commercial charging [6]–[9] and residential charging stations [15], [16]. Since the commercial charging stations have advantages of pooling the resources [29] and can invest in state-of-the-art charging technology over the residential charging stations, they have assumed center stage as far as charging solutions are concerned. Again, the optimal charging strategies proposed for commercial charging stations broadly fall into two categories. The first class proposes to control the arrival of EVs at the charging station mostly by controlling the pricing [5], [6]. The second class of policies proposes optimization algorithms to schedule the EV charging at the station for a given set of charging requests [29]. Ghosh and Aggarwal [6] and Yan *et al.* [10] have solved the scheduling problem to minimize the cost, reduce the system losses [25], provide ancillary services to the grid [11]–[13], reduce maximum demand [26], improve load factor [25], and maximize the social welfare [6].

Recently, car manufacturers Volkswagen [21], Tesla [22], and NIO [23] have introduced a mobile charging service to charge EVs in China on pilot basis. The service allows EV owners to order MCSs from the app. The global commercial EV charging station market size is projected to grow from 2 115 000 units by 2020 to 30 758 000 units by 2027, at a compound annual growth rate of 46.6% [4], [24], and out of the total charging station market, mobile charging market is projected to reach \$2.5 billion by 2027. The cost of charging the EV using a MCS is slightly more than charging at a commercial charging station, but it has its own advantages: 1) EV owners do not have to take their vehicles to the charging station and wait for charging; 2) EVs are charged at the convenience of the owner without having to spare time separately for charging (e.g., during the office hours); 3) charging stations also have the flexibility to schedule charging as per their convenience; and 4) the EV load gets decoupled from the distribution grid and can, thus, help flatten the load curve. In addition, the mobile charging service acts as an alternative to the fixed EV charging station, and thus, the choice lies with the EV owner to choose between the two depending upon their utility.

The problem of large-scale EV charging using MCSs has been addressed in [17]–[20]. Zhang *et al.* [19] have proposed to augment the existing charging infrastructure by deploying the mobile charging vehicle to locations where demand is expected to be high, while Chen *et al.* [18] propose a chaotic evolution particle swarm optimization algorithm to determine the optimal placement of MCSs based on expected demand. However, Chen *et al.* [18] and Zhang *et al.* [19] have not considered the stochastic travel times of charging vans. Furthermore, Chen *et al.* [18] and Zhang *et al.* [19] deal with the dispatching of a mobile charging truck to the areas of expected high demand, where EVs will, then, wait for charging in a queue. Chauhan and Gupta [17] have solved the problem of scheduling the MCSs to different fixed charging stations (FCSs) based on EV charging demands using heuristic methods. Chauhan and Gupta [17] have considered neither the charging window of requests nor the stochastic travel times of charging stations. Huang *et al.* [20] propose a nearest-job-next (NJN) algorithm to route a single mobile charging vehicle from one charging request to next; the charging requests are assumed to be homogeneous. However, in practice, the charging requests are spatially and temporally heterogeneous in nature, and thus, the NJN strategy of routing the MCS is not going to be optimal as far as cost is concerned.

D. Organization

The rest of this article is organized as follows. We introduce the modeling of various components in Section II. Then, in Section III, we formulate the complete optimization problem

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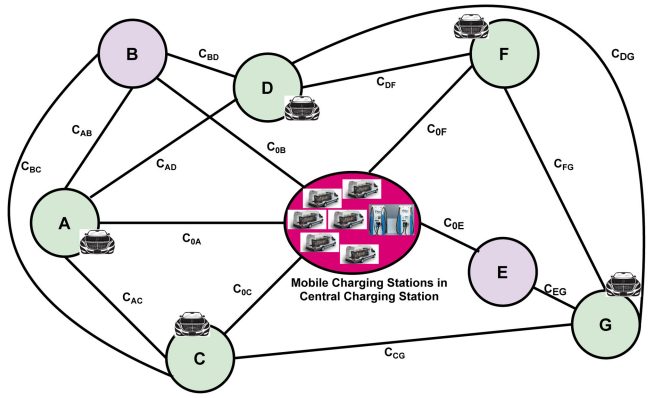

Fig. 1. Schematic to illustrate mobile charging service in a region consisting of seven areas A–G and a fleet of MCSs at the central charging station.

and present different solution methods to solve it in Section IV. Next, in Section V, we discuss the results of our simulations and show that our algorithms perform well in real-life situations. Finally, Section VI concludes this article.

II. Model

We consider a utility company/scheduling agency that receives the random charging requests (x, q, d) from the EV owners, where x denotes the position of the EV, q denotes the units of energy demanded, and d denotes the preferable time, by which the EV owners intend to receive the energy. Fig. 1 presents an example of mobile charging service in a particular area. Consider a particular scheduling horizon ΔT in which the utility company receives five charging requests from nonoverlapping regions A, C, D, F, and G. The job of the utility company is to assign the requests to the minimum number of MCSs such that the requests are serviced preferably before the deadline and battery capacity constraints of MCSs are not violated.

A. Heterogeneous EV Charging Requests

We consider a set of EVs $E = \{1, 2, \ldots, n\}$ that raised the charging requests during the preceding time interval ΔT, located at n different locations. Let x_i and x_j, respectively, be the position of the ith and jth EVs, where $i, j \in E$ such that $i \neq j$. We have assumed that the distances d_{ij} $\forall i, j \in E$ between any two pair of EVs are symmetric, i.e., $d_{ij} = d_{ji}$. However, the travel times t_{ij} are stochastic in nature due to changing traffic levels. Let the energy demand of the ith EV be represented by q_i located at position x_i. The charging requests of EVs represented by (x_i, q_i, d_i) received by the utility company are spatially and temporally random in nature. The spatial and temporal randomness of the charging requests along with random energy demands and stochastic travel times makes the problem of scheduling and routing of MCSs complex.

B. Service Area

The total geographical area from which charging requests are serviced is regarded as the service area. We discretize the service area into smaller area elements each represented by v_i, $i = \{1, 2, \ldots, a\}$, where a is the total number of segments the service area is divided into. It is needless to mention that the area elements can be chosen to be as small as desired, however at the cost of computational burden. The distance d_{ij} for MCSs between each pair of areas i and j, $\forall\, i, j \in \{1, 2, \ldots, a\}$, are assumed to be Euclidean. Thus

$$d_{ij} + d_{jk} \geq d_{ik}.$$

C. Mobile Charging Stations

Let $S = \{1, 2, \ldots, m\}$ be the set of MCSs that need to be routed and scheduled through m different routes to charge n EVs. We consider the MCSs to be homogeneous; thus, there is no precedence among them while assigning the charging requests. The capacity of each MCS $k \in S$ is represented by Q_k. We let $G_k = \{g_k(1), \ldots, g_k(n_k)\}$ denote the route of the kth MCS, $\forall k \in S$, where $g_k(j)$ represents the kth MCS servicing the jth EV and n_k is the total number of EVs being charged by the jth MCS. We assume a central charging facility located at origin (x_0), where MCSs return back after finishing the charging requests to recharge their batteries. The distances d_{0i}, $i \in E$, are measured with respect to the central charging station. All the routes begin and terminate at the central charging station x_0; thus

$$g_k(n_k + 1) = 0 \qquad \forall\, k \in S.$$

D. Charging Deadlines/Charging Windows

The charging request (x_i, q_i, d_i) raised by the owner of each EV $i \in E$ has a preferable deadline d_i by which the EV expects to receive the amount of energy q_i. Alternately, the EV owners may specify the time window, i.e., an interval of time within which the EV needs to be charged $[e_i, d_i]$, where $e_i < d_i$ represents the earliest time after which the vehicle is available for charging and the deadline or latest time by which charging should be completed for the EV $i \in E$. In addition, we represent the time required to complete the charging task of the vehicle $i \in E$ by t_{c_i}, which is proportional to energy demanded q_i.

E. Utility Company

The utility company has the distance and the expected travel time data available with it and has bidirectional communication with EVs and MCSs. On receiving the charging requests from EVs and the availability consent from MCSs, the utility company runs the optimization algorithm to route and schedule the MCSs. The optimization algorithm assigns different charging requests to MCSs and the sequence in which each MCS should service the EVs while minimizing the total cost of charging.

III. Problem Formulation

We formulate the problem as a constrained time and capacity graph-theoretic problem $\mathcal{G} = (N, A, c, T, t_i, t_f, q, Q)$ defined by the node set $N = E \cup \{0, n+1\}$, where $E = \{1, 2, \ldots, n\}$ is the set of EV locations and 0 and $n+1$ are the central charging facilities wherefrom the MCS starts and returns back

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after servicing the assigned charging requests, arc set $A = A(N)$ (the routes between the N nodes represent the road network), weights on the arcs $c(A) \in Z$ that represents the cost or the distance of that road, expected time taken to span arcs $T = T(A)$, the time of arrival and the deadline on nodes l and m where $t_i^l \geq t_i^0$ and $t_i^m \geq t_i^l + T_{lm}$ for $(l, m) \in N$, charging demands on nodes $q = q(N)$, and the capacity of mobile charging stations $Q = \{Q_1, Q_2, \ldots, Q_m\} \forall\, Q \geq q$. The problem is to determine the routes for m MCSs, where each route begins and terminates at the central charging station. Each MCS services a subset of total EVs to be charged in the particular scheduling horizon in a predetermined order and finally returns back to the central charging station x_0 to be ready for the next scheduling horizon.

Thus, the MCS routing problem is defined as follows. For a capacity and time-constrained digraph, find the m routes of the MCS of the minimum cost.

A. Scheduling and Routing With Hard Time Window Constraints

Definition 1: For every route G_k of MCS $k \in S$, we associate a binary decision variable defined as

$$x_{ij}^k = \begin{cases} 1, & \text{if } j \text{ is charged after } i \text{ by MCS } k \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $i \neq j \;\forall\; i, j \in N$.

We denote the cost of travel for MCS k from vehicle i to vehicle j $\forall i, j \in E$ as c_{ij}^k.

Definition 2: For every EV charging request $i \in E$ and MCS $k \in S$, we associate a binary decision variable defined as

$$y_i^k = \begin{cases} 1, & \text{if MCS } k \text{ charges EV } i \\ 0, & \text{otherwise} \end{cases}. \tag{2}$$

The subset of EV charging requests E should be assigned to each MCS $k \in S$ such that the sum of energy of charging requests assigned to MCS k should not exceed its maximum capacity Q_k; thus, we have the following theorem

Theorem 1: We say that the route G_k of MCS k is feasible in $\mathcal{G}$ if

$$\sum_{i=1}^{n} q_i \;\; y_i^k \leq Q_k, \qquad k = \{1, 2, \ldots, m\} \tag{3}$$

where q_i represents the energy demand of the ith EV and y_i^k is a decision variable that represents MCS k charging EV i.

Theorem 2: The following constraint ensures the elimination of split deliveries:

$$\sum_{k=1}^{m} y_i^k = \begin{cases} m, & i = 0 \\ 1, & i = \{1, 2, \ldots, n\} \end{cases}. \tag{4}$$

The above constraint ensures that of all the MCSs, only one visits each node $E = \{1, 2, \ldots, n\} \in N$ and the sum of all the MCSs that leave the central charging station is m. In addition to ensuring that each EV charging request is exactly on a single route, (4) also ensures that each route originates and terminates at the central charging station.

Once the EV charging requests are assigned to various MCSs, the next problem is equivalent to the routing problem with time window and capacity constraints.

Theorem 3: The following two constraints ensure that the m routes enter and leave the nodes $E \in \mathcal{G}$ exactly once

$$\sum_{i=0}^{n} x_{ij}^k = y_j^k \quad \begin{cases} j = \{1, 2, \ldots, n\} \\ k = \{1, 2, \ldots, m\} \end{cases} \tag{5}$$

$$\sum_{j=0}^{n} x_{ij}^k = y_i^k \quad \begin{cases} i = \{1, 2, \ldots, n\} \\ k = \{1, 2, \ldots, m\} \end{cases} \tag{6}$$

Equation (5) ensures that the kth MCS $\forall\, k \in S$ arrives at the EVs assigned to it only once, and (6) ensures that the kth MCS $\forall\, k \in S$ leaves from the EV only once.

In addition, the charging requests are time constrained, i.e., the problem at hand is the capacity-constrained MCS routing with the time window constraint.

Definition 3: Let e_i and d_i, respectively, represent the earliest and latest time during which EV $i \in E$ is available for charging.

Theorem 4: We say that the route $G_k = \{g_k(1), \ldots, g_k(n_k)\}$ $\forall\, k \in S$ is feasible if

$$e_i \leq t_i \leq d_i, \qquad i = \{1, 2, \ldots, n\} \tag{7}$$

$$t_i \geq 0, \qquad i = \{0, 1, 2, \ldots, n\} \tag{8}$$

where t_i is the time of arrival of MCS k at EV i.

In order to eliminate subtours, i.e., the MCS should not complete the assigned charging requests in the number of disjoint tours, we introduce the following constraint:

$$t_j \geq t_i + t_{c_i} + t_{ij} - (1 - x_{ij}^k) Z \quad \begin{cases} i = \{1, 2, \ldots, n\} \\ j = \{1, 2, \ldots, n\} \\ k = \{1, 2, \ldots, m\} \end{cases} \tag{9}$$

where t_{c_i} represents the time required to complete the charging task of vehicle $i \in E$. Equation (9) together with (10) and (11) ensures compatible arrival times for MCS k, $\forall\, k \in S$, at the assigned EVs, where Z is an arbitrary scalar constant whose magnitude is greater than the total scheduling horizon. Variables t_j and t_i, respectively, represent the arrival times of MCS k at the jth and ith EVs if the kth MCS travels to charge EV j after charging EV i

$$t_{0f}^k \geq t_i + t_{c_i} + t_{i0} - \left(1 - x_{i0}^k\right) Z \quad \begin{cases} i = \{1, 2, \ldots, n\} \\ k = \{1, 2, \ldots, m\} \end{cases} \tag{10}$$

$$t_j \geq t_{0s}^k + t_{0j} - \left(1 - x_{0j}^k\right) Z \quad \begin{cases} j = \{1, 2, \ldots, n\} \\ k = \{1, 2, \ldots, m\} \end{cases} \tag{11}$$

where t_{0f}^k and t_{0s}^k are the arrival and departure times of MCS $k \in S$ to and from the central charging station x_0, respectively. Equation (10) ensures that the MCS k arrives back at the central charging station only after servicing the last charging request in the route, while (11) ensures the time after which the MCS k can charge the first charging request on its route. However, the MCS is constrained not to leave the central charging station

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before the scheduling horizon and must return back by the end of scheduling horizon, so as to be fully charged back to service the charging requests for next scheduling horizon; thus

$$t_{0s}^k \geq e_0^k \qquad k = \{1, 2, \ldots, m\} \tag{12}$$

$$t_{0f}^k \leq d_0^k \qquad k = \{1, 2, \ldots, m\} \tag{13}$$

where e_0^k and d_0^k are, respectively, the earliest time and the latest time, by which MCS $k \in S$ can leave and return back to the central charging station.

After receiving EV charging requests during any time interval ΔT, the utility company runs an optimization algorithm at the end of time interval to clear the charging requests.

Definition 4: We say that only those charging requests can be serviced for which

$$d_i \geq T_{\min}, \qquad i \in \{1, 2, \ldots, n\}$$

where $T_{\min}$ is the minimum deadline set by the utility company for the EV charging requests and d_i is the deadline of charging request $\forall\, i \in E$.

This is to take into account the travel time of MCSs and the time taken to charge the battery of EVs. The job of the utility company, thus, is to assign the received charging requests to different MCSs and assign the order in which the charging requests should be serviced to solve the following optimization problem:

$$\begin{aligned} &\text{minimize} \quad \sum_{i=0}^{n} \sum_{j=0}^{n} \sum_{k=1}^{m} c_{ij}^k \;\; x_{ij}^k \\ &\qquad\qquad \text{subject to (1)–(13)} \\ &\text{variables}: \;\; x_{ij}^k, \;\; y_i^k, \;\; t_i, \;\; t_{0s}^k, \;\; t_{0f}^k. \end{aligned} \tag{14}$$

B. Scheduling and Routing With Time Windows in the Presence of Stochastic Travel Times of MCSs

So far in the modeling, we have considered the time window at the nodes a hard constraint. However, in real life, the travel time for MCSs from one service request to another is stochastic in nature. The MCSs have to operate on the traffic network, which has different levels of congestion at various intervals of the day and, thus, stochastic travel times between the nodes. Thus, it becomes a very strong assumption to schedule and route the MCS with respect to a deterministic and static schedule. In the presence of stochastic travel times, the scheduling and routing of MCSs to service EV charging requests considering hard time constraints will result in unrealistic and inefficient scheduling. We, thus, modify the optimization problem in (14) to incorporate the hard time window constraints as soft constraints. The constraint (9) ensures the compatible arrival times between any two consecutive charging requests i and j. Constraints (10) and (11) are explicitly for the central charging station, while constraint (7) enforces the time window constraints on the servicing of charging requests.

In the optimization problem (14), the time window constraint constitutes a hard constraint and, thus, should not be violated in any feasible solution. In order to transform the hard time constraints into soft time constraints (which does not necessitate the 100% on-time service), we introduce the constraint (7) using Lagrangian relaxation into the objective function multiplied by a suitable penalty factor.

C. Final Optimization Problem

Then, the overall objective function becomes

$$\mathcal{P} = \sum_{i=0}^{n} \sum_{j=0}^{n} \sum_{k=1}^{m} w_D \;\; c_{ij}^k \;\; x_{ij}^k + \; w_T \;\; u_{ij} \;\; x_{ij}^k \tag{15}$$

where

$$\begin{aligned} u_{ij} = {} & 1/2\,[p_e \max(0, e_i - t_i) + p_l \max(0, t_i - d_i)] \\ & + 1/2\,[p_e \max(0, e_j - t_j) + p_l \max(0, t_j - d_j)] \end{aligned}$$

where w_D and w_T are the weighing factors corresponding to distance and time, respectively and c_{ij}^k is the cost matrix corresponding the Euclidean distance between nodes i and j. The term u_{ij} measures the penalty due to the violation of time windows at the end of each link (i, j). The factor $1/2$ ensures that the penalty is counted only once per violation. Again, p_e and p_l are the early charging service penalty and the late charging service penalty, respectively. e_i and d_i are the earliest time and the latest time, respectively, during which charging is preferred by the EV owner. The max function will ensure that the penalty is imposed only when charging request is serviced outside the preferred time slot. The two terms in the overall cost function correspond to the spatial component (routing) and the temporal component (scheduling), respectively.

Then, the final optimization problem becomes

$$\begin{aligned} &\text{minimize} \quad \mathcal{P} \\ &\text{subject to} \;\; \text{(1)–(6) and (8)–(13)} \\ &\text{variables}: x_{ij}^k, \;\; y_i^k, \;\; t_i, \;\; t_{0s}^k, \;\; t_{0f}^k. \end{aligned} \tag{16}$$

D. Effect of Considering the FCS Along With MCSs

In a practical situation, the EV owners will have both the options available to either get their EVs charged at an FCS (commercial charging station) or request for an MCS. Depending on the utility of the EV owner and the feasibility of her charging request, she may prefer to get her vehicle charged at an FCS if the waiting time is less at the FCS or can request the MCS. Thus, on raising the charging request, the EV owner gets the information of charger availability and waiting time at the FCS, and as per utility, she decides to either book charger at the FCS or request a mobile charging service within a particular time interval.

Thus, of the total charging requests, a subset of requests will be scheduled to the FCS and only a subset of requests will be serviced using MCSs. The MCS operator will then solve the optimization problem in (16) to schedule and route the MCSs to service the subset of requests using MCSs. This will substantially reduce the number of MCSs and the cost of delivering the energy to EVs as the subset of total requests will be serviced at the FCS.

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IV. Solution Technique

The optimization problem in (16) is equivalent to the classic capacitated vehicle routing and scheduling problem with time windows (CVRSPTW) [28]. This problem is a combination of two NP-hard problems: the traveling salesman problem and the bin packing problem. This is the classic case of the NP-hard combinatorial optimization problem and, thus, cannot be solved in polynomial time for the large-scale scheduling and routing of MCSs. However, for smaller instances, the problem can be solved optimally using conventional techniques. We present three different methods to solve the optimization problem in (16) and evaluate them on the following metrics.

1) *Computation complexity:* This metric is related to the computational complexity of the method. Thus, even if the method guarantees the optimal solution but is computationally complex, it may not be useful.
2) *Scalability:* This metric reflects the ability of the method to be equally efficient as the size of the problem increases, i.e., as the number of charging requests and MCSs increase.
3) *Quality of the solution:* This metric reflects the ability of the method to provide the optimal solution or the solution that is near optimal.

A. Exact Algorithm Mixed-Integer-Linear-Programming (MILP) Solution

We solve the MILP optimization problem in (16) for small instances using CVX package and Gurobi solver [32]. The advantage with the exact algorithm is the guarantee of the optimal solution by exploration of the whole search space. The exact method is computationally intensive, and thus, the combinatorial optimization problem in (16) cannot be solved in polynomial time as the size of the problem increases.

B. Heuristic Methods

In contrast to exact algorithms, the heuristic algorithms explore only a subset of total search space; thus, they have the advantage of giving reasonably good solutions in small time. Though heuristics methods do not guarantee optimal solutions, they provide solutions fairly quickly and are computationally efficient. Heuristics are scalable and are well suited to solve a large-sized problem but at the cost of the quality of solution.

We propose the modified Clarke and Wright's savings heuristic [31] to solve the optimization problem in (16) and compare its results with the benchmark NJN heuristic. "Savings" heuristic remains by far the best-known heuristics that is used to solve the CVRSPTW problem for its speed and simplicity. The heuristics cannot guarantee the optimal solution, but the savings algorithm has been used extensively in operations research to solve such problems with near optimal results in the reasonable time [28]. This heuristic is based on the concept of savings; let i and j be two EV charging requests at distance $X_{0i} = 15$ km and $X_{0j} = 18$ km from the central charging station, respectively, and $X_{ij} = 16$ km from each other, scheduled to be serviced by two different MCSs. The total distance to be covered by two MCSs is $\{2(X_{0i} + X_{0j})\} = 2 \times (15 + 18) = 66$ km. Next, if it is feasible to service both the charging requests by a single MCS, then the total distance covered is $\{X_{0i} + X_{ij} + X_{j0}\} = 15 + 16 + 18 = 49$ km. As distances are assumed to be Euclidean in nature, the savings by combining the two service requests into one will yield savings S_{ij} of

$$S_{ij} = \{2(X_{0i} + X_{0j})\} - \{X_{0i} + X_{ij} + X_{j0}\}$$
$$S_{ij} = X_{0i} + X_{0j} - X_{ij}$$
$$S_{ij} = 17 \text{ km}.$$

Algorithm 1: Modified Saving's Algorithm.

1: At the beginning of the scheduling horizon, the savings are calculated for each pair (i, j) of charging requests, which are then arranged in the form of a list S in decreasing order.
2: Starting from the first entry which gives the highest savings a route is built connecting service request i and j provided the charging time deadlines of EVs and capacity constraints of the MCSs are not violated.
3: If however, either ith or jth service requests are already assigned to any other route, then the arc X_{ij} is connected to that route provided either i or j request lies at the external point of the partial route.
4: In case any of the deadline constraint or capacity constraint violations are found then a new route containing X_{ij} is initiated.
5: Similarly if both ith and jth charging requests are already assigned to two different routes, and both the charging requests are the exterior points, then the two routes are merged provided no time or capacity constraints are violated.
6: The savings list is updated by removing the term S_{ij}. If the savings list is not exhausted, go to step 2, else end.

Algorithm 1 describes the steps of modified savings heuristics to solve the optimization problem in (16).

C. Metaheuristics Methods

Metaheuristics have been widely used to solve the combinatorial optimization problem in literature. Metaheuristics provide near-optimal solutions but are faster than the exact methods. They are, thus, scalable and provide good-quality solutions in reasonable time. We propose a modified genetic algorithm (GA) [33] to solve the optimization problem in (16). The GA is a well-known effective population-based algorithm. GAs are based on the idea of natural selection and are capable of finding the global optimal or near-global-optimal solutions with fast convergence.

Since GAs by nature are better suited to solve problems of discrete nature, they are, thus, preferred for solving combinatorial optimization problems. Algorithm 2 presents the steps of the modified GA. The steps of mutation and crossover are general in nature, while the steps of chromosome structure coding and fitness function evaluation are problem specific.

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1) Chromosome Structure Coding: The structure of each chromosome in the GA represents the possible solution. As we need to find m routes of MCSs serving n charging requests, the coding structure of chromosome is

$$\left(\overbrace{0, E1_1, E2_1, \ldots, El_1}, \overbrace{0, E1_2, E2_2, \ldots, El_2}, \ldots \overbrace{0, E1_m, E2_m, \ldots, El_m}\right)$$

where 0 represents the central charging station, and each group represents the charging requests serviced by m MCSs in sequence. Care should be exercised that no charging request occurs more than once on the chromosomes.

2) Generation of Initial Population: The set of initial chromosomes that are generated randomly are called population. The initial population may by generated randomly or using any sweep heuristics. In the sweep method, the total charging requests are divided into m groups such that all the constraints are met.

3) Fitness Function: In order to determine the quality of various individuals (solutions), they are evaluated using the fitness function defined as

$$\mathcal{Z} = \left\{\sum_{i=0}^{n}\sum_{j=0}^{n}\sum_{k=1}^{m} w_D \; c_{ij}^k \; x_{ij}^k + w_T \; u_{ij} \; x_{ij}^k\right\} + w_c \sum_{k=1}^{m} \max\left\{\left(\sum_{i=1}^{n} q_i \, y_i^k - Q_k\right), 0\right\}. \quad (17)$$

4) Natural Selection: The natural selection is the process of selecting the best chromosomes or solutions from the population by evaluating them using the fitness function. The fittest individuals then produce the offspring, which have characteristics of the parents, and subsequently, generation with fittest individuals will be found following the notion of survival of fittest.

5) Crossover: Crossover is the most important part of the GAs. Each of the two chromosomes (parents) that mate contribute a part of their gene sequence in the offspring. However, to ensure that the chromosomes after crossover (offspring) do not become invalid, we develop a method of modified crossover.

Let $p1 = \{\overbrace{0, 4, 6, 2, 1}, \overbrace{0, 3, 5}\}$ and $p2 = \{\overbrace{0, 5, 2, 1}, \overbrace{0, 6, 3, 4}\}$ be the two parent chromosomes; then, at the crossover say at gene 1 in $p1$, $[1, 3, 5]$ are incorporated into $p2$ to produce offspring. The elements $(1, 3, 5)$ already present in $p2$ are first removed from it. Thus, the resulting offspring remains a valid solution $O1 = \left\{\overbrace{0, 2}, \overbrace{0, 6, 4, 1, 3, 5}\right\}$.

6) Mutation: In this article, we present two different methods of mutation: inversion and exchange. For example, the portion of chromosome $(4, 3, 1)$ can invert to $(1, 3, 4)$ or it can shift ahead resulting in

$$\left\{\overbrace{0, 2}, \overbrace{0, 6, 1, 3, 4, 5}\right\} \text{ and } \left\{\overbrace{0, 2, 1, 3, 4}, \overbrace{0, 6, 5}\right\}.$$

Algorithm 2: Modified GA.

1: **Initialization** At the beginning, initial population of chromosomes are randomly generated.
2: **Fitness Evaluation** The fitness of solutions is evaluated and only the best individuals are selected for reproduction.
3: **Termination** If the solutions have converged, the algorithm is terminated and the individual with best fitness is proposed as the solution.
4: **Crossover** During the mating of two parents, the offspring will have features of both the parents.
5: **Mutation** In the offspring, certain genes are subject to mutation to prevent premature convergence due to local minima.
6: Back to step 2.

V. Simulation Results

We numerically study and evaluate the various scheduling algorithms present in this article and show that the cost of charging, the number of MCSs, and the miss ratio are reduced in the proposed algorithm. We also show that the number of MCSs required and the cost of delivering unit energy are less if FCSs are used along with MCSs. We compare the results of three solution methods, i.e., 1) exact, 2) heuristics, and 3) metaheuristics, on the metrics of computational complexity, scalability, and quality of solution.

A. Simulation Setup

To test our algorithms, we randomly generate charging requests from different area elements for each interval of time ΔT. At the end of each time interval ΔT, the utility company runs an optimization algorithm to dispatch MCSs to the charging requests. For an average energy consumption of 0.15 kWh/km and considering that in one single journey to office, a person normally travels between 40 and 50 km and would, thus, normally require the battery energy of 7–10 kWh. Thus, similar to the "Experiences from six smart grid investment grant projects" [30], we take the average amount of energy demanded per EV as 6.9 kWh with standard deviation 4.9 kWh in the interval [2, 20]. In addition, from [30], we take the deadline of charging requests to be exponentially distributed with mean 2.5 h. In simulations, we study the effect of the mean deadline (1.5–5.5 h) of charging requests on the scheduling and number of MCSs required. The arrival process of EV charging requests is considered to be a nonhomogeneous Poisson process, and we study the effect of charging requests received per hour on the scheduling in simulations. We divide the total service area (20 km $\times$ 20 km) 400 km^2 around the fixed central charging station into smaller area elements of 1 km^2 each. We assume that the FCS has 20 charging slots and the EV owner will prefer fixed chargers over MCSs if there is no waiting time. In the simulation process, we assume that EV owners will prefer FCS until all the chargers are occupied.

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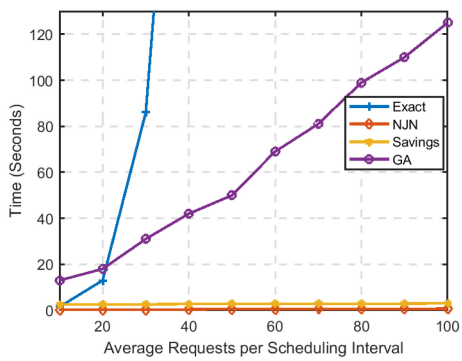

Fig. 2. Comparison of computation time for different solution methods.

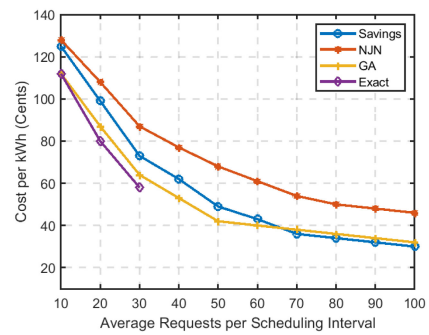

Fig. 3. Comparison of average cost per unit charge in different methods considering stochastic travel times of MCSs.

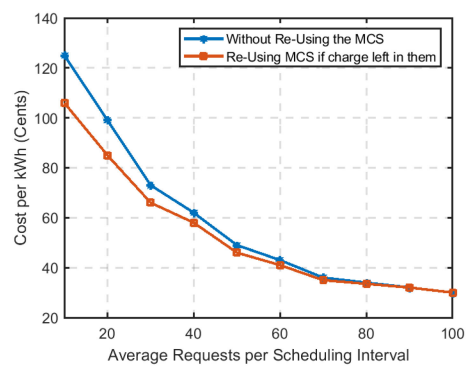

Fig. 4. Average cost per unit energy with and without reusing the MCSs.

The distance of each area element from the central charging station and the distance between each pair of area elements are available with the utility company. The expected travel time of MCSs to travel between different area elements is also updated at the beginning of each interval. MCSs carry fast dc–dc chargers capable of charging at 50-kW power and carrying a battery bank of 100-kWh capacity. We also study the effect of varying the capacity of the MCS on the scheduling and routing of MCSs.

In the heuristic method, we compare the proposed savings heuristic with the NJN heuristic. The NJN policy chooses to service the nearest request first and then the request that is nearest to the previously serviced request. In the GA metaheuristic, we tune the parameters by performing the sensitivity analysis by varying one parameter at a time and keeping other parameters fixed. We choose the spatial penalty factors $w_D = 2000$, temporal penalty factor $w_T = 3000$, and the penalty corresponding to exceeding the capacity of MCSs $w_c = 20000$. We choose the early charging penalty $p_e = 200$ and the late charging service penalty $p_l = 200$. Again, we choose the population size as 100, maximum iterations or generations as 400, the crossover rate as 0.85, and the mutation rate as 0.2.

B. Comparison of Three Solution Methods

Fig. 2 shows the comparison of time taken by different methods to arrive at the acceptable solution. As evident from Fig. 2, the time required by both the heuristics is small and remains almost the same as the size of the problem increases. In contrast, the time taken by the GA increases with the increase in the size of the problem. Fig. 2 shows that the time taken by exact methods increases exponentially as the size of the problem increases, and thus, the optimization problem in (16) can be solved by exact methods only for small instances.

Fig. 3 shows the comparison of the quality of solution by different methods. For small instances, the exact algorithms give the optimal solution followed by the GA, savings heuristic, and NJN heuristic. However, for the medium-sized problem, exact methods cannot solve the combinatorial optimization in reasonable CPU time, and thus, the quality of solution by the GA is better than the savings heuristic, which is better than the NJN. Finally, for the large-sized problem, it is evident from the figure that savings heuristics perform slightly better than the GA. This is due to GA's randomness of search direction in big solution spaces. Thus, for large-sized problems, the modified savings heuristic performs better than the GA in terms of not only optimal results but also the shorter computing time. The savings algorithm can be further improved (see Fig. 4) to reduce the cost if the EV charging requests that are received after the MCSs have left are assigned to the already servicing MCS if they have energy left and time constraints are not violated. However, the MCS can be reused only if they have capacity left at the end of route, but as the number of charging requests per interval increases, most of the MCSs are assigned requests up to their capacity. Thus, only a few MCS are available to be reused.

Thus, for small-sized problems, exact algorithms are best; for medium-sized problems, the GA is better; and for the large-scale problem, modified savings heuristic performs the best.

C. Theoretical Bounds on the Optimal Solution

For any heuristic method, the theoretical bounds on the optimal solution are basic tool for evaluating the quality of solution from the heuristic. It gives the guarantee that the solution will lie between the lower bound and the upper bound. In general, the lower bounds are obtained by solving the relaxations of the original problem. In this article, we have used the method of spanning trees by removing one edge from any Hamiltonian cycle to find the lower bound of the routing problem of MCSs . Thus, we have the following.

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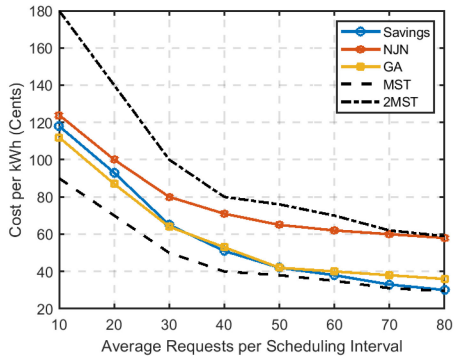

Fig. 5. Comparison of the average cost of charging an EV per kilowatthour in the proposed algorithm and the NJN policy.

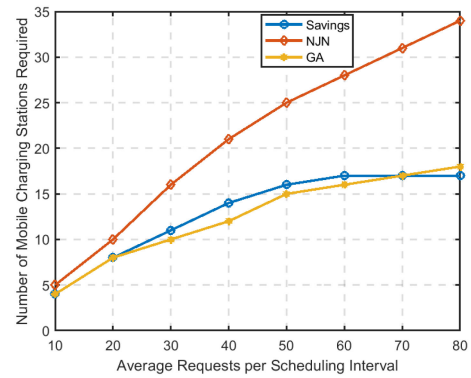

Fig. 6. Comparison of the number of MCSs required with different rates of charging requests in the proposed algorithm and the NJN policy.

1) *Solution of the routing problem > minimum spanning tree (MST):* Since the distances are Euclidean in nature and

$$d_{ij} + d_{jk} \geq d_{ik}$$

the upper bound on the routing problem of MCSs can be obtained by constructing a Hamiltonian cycle from minimum spanning tree and replacing edges with shortcuts.

2) *Solution of the routing problem < 2MST:* Fig. 5 shows the cost per unit of energy in the savings heuristic along with the upper and lower bounds.

D. Results

In this subsection, we compare our scheduling and routing algorithm to the greedy NJN policy. Furthermore, we study the effect of mean deadlines, mean energy demands, and scheduling frequency on the scheduling and routing policy of MCSs.

1) Cost of Charging: Fig. 5 shows that the average cost of charging an EV using the proposed optimization algorithm is considerably less than that in the greedy NJN policy. The advantage is particularly large when demand is high because MCSs are busy, and thus, optimization helps reduce the cost significantly. In contrast, at low demands, as most of the MCSs are underloaded, the greedy policy performs as good as the optimization algorithm for the same number of MCSs.

2) Number of MCSs: At low demands, i.e., for less frequency of charging requests, Fig. 6 shows that the number of MCSs required is similar in both the algorithms. However, at large frequency of charging requests, the algorithm makes sure that more requests are serviced using the existing MCSs, and thus, our algorithm requires less number of charging stations.

3) Miss Ratio: The miss ratio is the percentage of charging requests missed to the charging requests received. Fig. 6 illustrates that for the same number of MCSs, e.g., 15, our algorithm optimizes the scheduling and routing considering the time and capacity constraints and, thus, charges the maximum number of EVs (i.e., 45) compared to the greedy NJN policy (28), which is greedy and charges the EVs in its neighborhood, thus making it suboptimal.

4) Effect of Average Deadlines: As the average deadline of charging requests increases, Fig. 7 shows that the number of MCSs required reduces at a faster rate initially as more charging requests can be serviced using the same number of MCSs. However, the curve flattens as the deadline of charging requests increases further because less and less charging requests can be serviced using the existing MCSs as they are almost servicing requests up to their battery capacity.

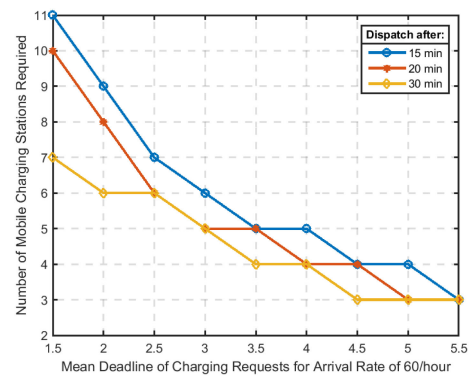

Fig. 7. Number of MCSs required with the mean deadline of charging requests for different frequencies of scheduling.

5) Effect of Scheduling Horizon, i.e., Frequency of Dispatch: Fig. 7 shows that the number of MCSs required and, thus, the average cost of charging each vehicle decrease or the percentage of extra profit increases with the decreasing frequency of dispatch. This is because as the time between two dispatch instances increases, it gives the additional time to dispatch the MCSs to charging requests and, thus, make better decisions.

6) Effect of Minimum Prior Charging Request Time: The prior charging request time is the minimum time between the instant the charging request is raised and the deadline by which the EV owner wants her vehicle charged. Similar to increasing the mean deadline (see Fig. 7), this policy parameter $T_{\min}$ has an impact on the routing policy. It is clearly evident that the increase in the minimum prior charging request time gives additional time to make optimal decisions and, thus, reduces the cost and the number of MCSs.

7) Effect of Frequency of Charging Requests: Fig. 5 shows that as the frequency of charging requests increases, the cost of charging initially decreases fairly steeply, but afterward, the

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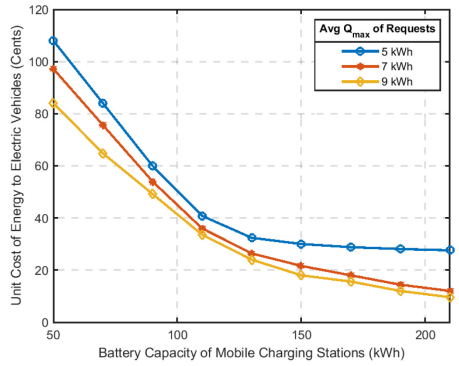

Fig. 8. Comparison of the average cost of charging an EV per kilowatthour in the proposed algorithm and the NJN policy.

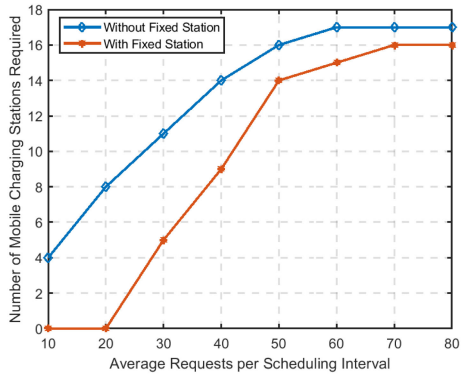

Fig. 9. Comparison of the number of MCSs required with and without considering FCSs in the simulation process.

curve becomes less steep and finally flattens. This is because initially as the frequency of charging requests increases, more charging requests are serviced by a particular MCS as more charging requests are available, but as the frequency of charging requests increases further, a particular MCS cannot take any more charging requests because of its battery capacity limit.

8) Effect of the Maximum Battery Capacity of the MCS: Initially, it seems that the higher the storage capacity of MCSs, the lower the overall cost as less number of MCSs would be required, but Fig. 8 shows that as the number of charging requests serviced by each MCS increases, it is most likely to miss the deadline of the charging requests that are scheduled for the later part of route. Thus, the marginal increase in storage capacity does not have much value after a particular value of storage capacity though the capital cost increases linearly.

9) Effect of Magnitude of Energy Required Per Charging Request: As the mean value of energy demanded $q_{\max}$ increases, less number of charging requests can be assigned per MCS, i.e., MCSs are utilized completely even at moderate demands. Fig. 8 shows that the cost of delivering per unit energy to EV decreases as the average energy demanded by charging requests increases.

10) Effect of Considering the FCS Along With MCSs: In the scenario of considering both mobile and fixed charging together, we assume that, initially, all the fixed chargers will be occupied and then MCSs. Fig. 9 shows that the number of MCSs required is reduced as the subset of charging requests received will be

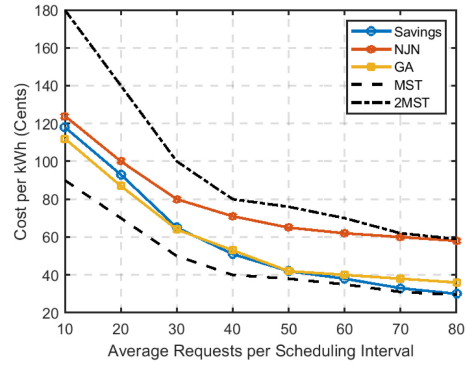

Fig. 10. Comparison of the average cost of charging an EV per kilowatthour with and without considering FCSs.

serviced at the FCS. Again, as seen from Fig. 10, the cost of delivering per unit energy is also reduced when fixed chargers are available along with MCSs.

VI. Conclusion

In this article, we proposed an alternative charging strategy of MCSs to charge EVs without the constraints of time and location. The MCSs need to be scheduled and routed to various charging requests to minimize the cost and the number of MCSs required. We formulated the complete optimization problem and showed it to be a combination of a bin packing problem and a traveling salesman problem, which are NP-hard. We presented three different methods, i.e., exact, heuristic, and metaheuristic, to solve the optimization problem. Finally, using numerical simulations, we discussed how different parameters affect the performance of MCSs.

As a future work, the scheduling and routing of MCS as an online optimization problem is an interesting future direction. We consider in future to study the effect to deadline-differentiated pricing and menu-based pricing on the mobile charging service. In addition, the competition among various mobile charging service providers constitutes an important future research direction.

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